

Amrith Akshintala

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Portfolio: codingmastermind123.github.io/MainPortfolio/

EDUCATION

Texas A&M University

College Station, TX

Bachelor of Science in Computer Engineering

Expected May 2029

- **Honors:** Engineering Honors; Eagle Scout; AP Scholar with Distinction; National Merit Commended Scholar; PVSA Gold Award
- **Relevant Coursework:** Programming Design and Concepts, Multivariable Calculus, Electricity & Magnetism, Calculus II
- **Organizations:** Texas A&M TURTLE Robotics (foundational skills in Python, C++, electronics, and CAD modeling), Aggie Coding Club, Aggie Data Science Club

EXPERIENCE

Research Intern

Aug. 2025 – April 2026

Texas A&M Ultrasound Research Laboratory

College Station, TX

- Developed non-invasive ultrasound techniques to measure tumor pressure
- Created CAD models for treatment prototypes

Research Intern

June 2024 – Aug. 2024

Dartmouth-Hitchcock Medical Center

Lebanon, NH

- Developed computational models for cancer detection from pathological image data through Dartmouth Health's EDIT AI Program, a mentored research initiative in the Department of Pathology
- Collaborated within a research team under faculty mentorship, following IRB-approved protocols and presenting progress through a weekly seminar series

Co-founder

July 2023 – July 2025

Scholar Sphere

Remote

- Started a nonprofit providing online educational resources to high school students, including a full-length SAT practice platform built with HTML, CSS, and JavaScript
- Established partnerships with 22 nonprofits; earned the PVSA Gold Award for volunteer service

PROJECTS

2-Axis Gimbal | *Arduino, C++, GY-87 IMU, SG90 Servos, SolidWorks*

May 2026 – Present

- Designing a 2-axis camera gimbal with a team of 4 using an Arduino GY-87 IMU and SG90 servos; implemented PID stabilization in C++
- Completed full CAD modeling and assembly in SolidWorks; currently in final wiring and hardware assembly

Sensor Dashboard | *React, Node.js, Express, WebSocket, Recharts*

July 2026 – Present

- Built a full-stack real-time telemetry dashboard for embedded sensor data, with a Node.js/serialport bridge streaming Arduino Uno R4 data through an Express + WebSocket backend to a React/Vite frontend
- Implemented live charting with Recharts, a drag-and-resize dashboard layout using react-grid-layout, and CSV session logging via a custom recorder module

StudyBot | *React, Node/Express, Supabase, Claude API*

June 2026 – Present

- Built and deployed a full-stack AI study assistant with Google OAuth, row-level security, an interactive quiz system, and prompt caching (~90% cost reduction)
- Engineered a RAG pipeline with semantic vector search using OpenAI embeddings and Supabase pgvector

Disease Classification from Medical Images | *PyTorch, NumPy, Pandas*

Jan. 2026 – April 2026

- Built a ResNet-50 deep learning model to classify chest X-rays into 3 categories (Normal, Bacterial Pneumonia, Viral Pneumonia), achieving 86.4% validation accuracy
- Performed data preprocessing, augmentation, and evaluation using PyTorch, NumPy, and Pandas with the Aggie Data Science Club

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS, Swift, SystemVerilog

Frameworks & Tools: React, TensorFlow, PyTorch, SolidWorks

Professional: Leadership, Teamwork, Time Management, Problem-Solving, Communication